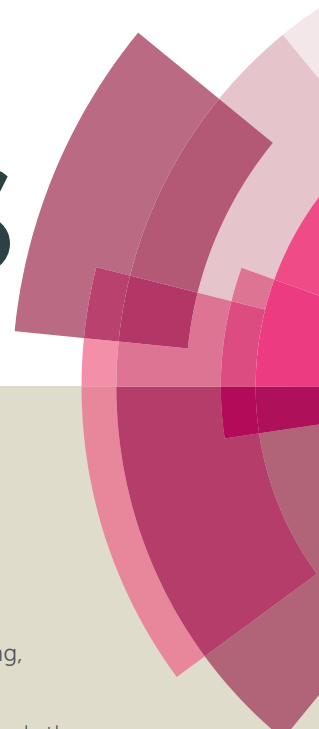


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COMMUNICATION

Enhanced Cyclability of Rechargeable Li-O₂ Batteries
Enabled by Boron Carbide

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A Li-O₂ battery with boron carbide (B₄C) based electrode presented a discharge-recharge voltage gap of 1.11 V, which is 550 mV smaller than that of Super P based Li-O₂ battery. The B₄C based electrode also enables much enhanced cyclability of over 160 cycles at a current rate of 1000 mA g⁻¹ (0.3 mA cm⁻²). The improved round-trip efficiency and cyclability is attributed to the high electrocatalytic activity and excellent stability of B₄C.

The rechargeable Li-O₂ battery is the most promising energy storage technology due to it extremely high theoretical specific energy of 3600 Wh kg⁻¹ compared with other energy storage systems.^[1] However, several challenges including high overpotential and inferior cyclability need to be addressed before they can be considered for future practical applications. High overpotential resulted from sluggish kinetics of cathode reaction causes large energy loss, poor round-trip efficiency and rate capability. Inferior cyclability caused by parasitic reactions between battery components and reaction species leads to premature cell death.^[2, 3] It is believed that the overpotential can be considerably reduced and the cyclability can be improved by developing stable and effective cathode catalysts that can enhance the reversible formation/decomposition of Li₂O₂ during discharge and charge and inhibit the occurrence of parasitic reaction.

Recent research effort focuses on searching stable materials such as nanoporous gold (Au),^[4] perovskite,^[5] titanium carbide (TiC),^[6] Sb-doped tin oxide (STO)^[7] and Magnéli oxide (Ti₄O₇).^[8] While the improved cyclability has been demonstrated with these materials, applying them in Li-O₂ batteries leads to significant trade-offs in the specific energy (mAh g⁻¹) and cost due to the higher density and price of those materials.^[9] Therefore, it is still one of the greatest challenges to identify highly stable and effective bifunctional catalysts that significantly increase the energy

efficiency and cyclability of Li-O₂ batteries. Here, for the first time we report that a cathode based on B₄C can overcome the disadvantages as mentioned above. It greatly reduces side reactions at the cathode/electrolyte interface due to electrode and electrolyte decomposition and is more stable and lighter than other non-carbonaceous materials (2.52 g cm⁻³ for B₄C compare to 4.93 g cm⁻³ for TiC and 19.32 g cm⁻³ for Au).^[10-14] Results demonstrate that the use of B₄C based cathode led to reduced polarization (ca. 1.1V), improved round-trip efficiency (71.2%) and stable cyclability that exceeded 50 and 160 cycles with the capacity limit of 5000 and 1000 mAh g⁻¹, respectively. The improved performance can be mainly attributed to the excellent stability and catalytic activity of B₄C. The possible oxygen electrode reaction mechanism is proposed on the basis of morphology and structural investigation.

Transmission electron microscopy (TEM) (Fig. 1a) and scanning electron microscopy (SEM) (Fig. S1) images clearly demonstrate the porous morphology of the sample, which comprises B₄C nanoparticles with a size of about 200 nm composed of small particles (~50 nm). In the HRTEM image (Fig. 1b), the measured distance between adjacent lattice fringes is about 0.24 nm, in accordance with the (021) plane of B₄C. The characteristic diffraction spots of corresponding SAED pattern shown in inset of Fig. 1b can also be assigned to the (021) facet of B₄C. The powder X-ray diffraction (XRD) patterns of B₄C powder showing in Fig. 1c demonstrates that the peaks can be indexed well to the well-crystallized B₄C (PDF no. 35-0798). The impurity peak at 34° is assigned to B (PDF no. 11-0617) and the peaks at 40° and 43° are assigned to B₂O₃ (PDF no. 44-1085). The N₂ adsorption-desorption plot of B₄C (Fig. 1d) shows a type IV isotherm with a steep increase of nitrogen absorption at a high relative pressure (P/P₀=0.85-0.99), indicating that the main pore volume is contributed by large size pores. The B₄C exhibited a high specific surface area of 52.2 m² g⁻¹ and the pore sizes are mainly distributed at about 4 and 100 nm (inset in Figure 1d). The high specific surface area would provide not only large electrolyte-electrode interface to ensure favourable accessibility of the active sites but also adequate storage space for

insoluble discharge products. The favourable pore structure would certainly facilitate the diffusion of reactants, ensure high availability of the catalytic active sites and thus certainly benefit the electrochemical performance of Li-O₂ batteries (vide infra).

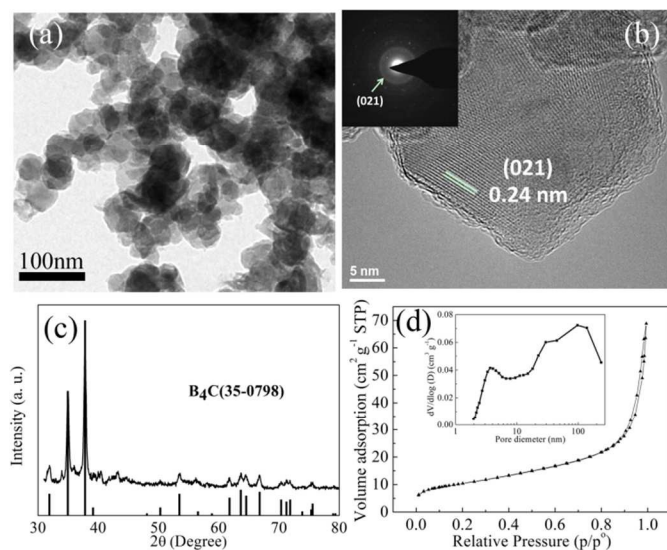


Fig. 1 (a) TEM image and (b) HRTEM image of the B₄C nanoparticles, and inset in (b) is the SAED pattern of B₄C; (c) XRD pattern of the B₄C nanoparticles; (d) N₂ adsorption-desorption isotherms, and inset is the pore-size distribution of B₄C nanoparticles.

Before investigate the electrochemical performance of B₄C based electrode, the effects of the substrate (carbon paper) should be excluded. The details can be found in Fig. S2-S5. It is obvious that the influence of carbon paper on the performance of Li-O₂ battery can be neglected. The initial galvanostatic discharge/charge curves of Li-O₂ batteries with B₄C and Super P cathodes at current density of 500 mA g⁻¹ (0.15 mA cm⁻²) are compared in Fig. 2a. A remarkably lower overpotential (1.11 V) and corresponding higher round-trip efficiency (the ratio of discharge to charge voltage) (71.2%) with a substantially larger specific capacity (6621 mAh g⁻¹, ~2 mAh cm⁻²) were demonstrated for B₄C electrode compared to Super P electrode (overpotential, round-trip efficiency and specific capacity of Super P based Li-O₂ batteries are 1.66 V, 61% and ~4800 mAh g⁻¹ (~1.44 mAh cm⁻²), respectively.), indicating that B₄C based cathode can effectively promote the electrode reaction and result in the improved performance of the Li-O₂ cell. During charge, OER achieves 100% completion below 4.0V which is far below the decomposition voltage of electrolyte (Figure S6). With current density increasing from 500 mA g⁻¹ (0.15 mA cm⁻²) to 1000 mA g⁻¹ (0.3 mA cm⁻²), the overpotential increases from 1.11 V to 1.52 V and the capacity decreases from 6621 mAh g⁻¹ (2 mAh cm⁻²) to 4040 mAh g⁻¹ (1.2 mAh cm⁻²). However, even at two times higher current density, the polarization of the B₄C based Li-O₂ battery was lower than that of Super P based battery. Our results thus demonstrate the improved rate capability of B₄C based Li-O₂ battery. Electrochemical impedance spectroscopy (EIS) measurements were conducted to study the electrochemical polarization of B₄C and Super P electrodes (Fig. S7, Tab. S1 and S2). After discharging, the charge transfer resistance (R_{ct}) of O₂ reduction on both B₄C and Super P are increased, implying that the formation of discharge products prevents reactant from diffusing to the surface of cathode and finally reduces the kinetics of oxygen reduction.^[15] Obviously, the charge transfer resistance (R_{ct}) of O₂ reduction on B₄C in the pristine, discharged, and recharged states are significantly lower than

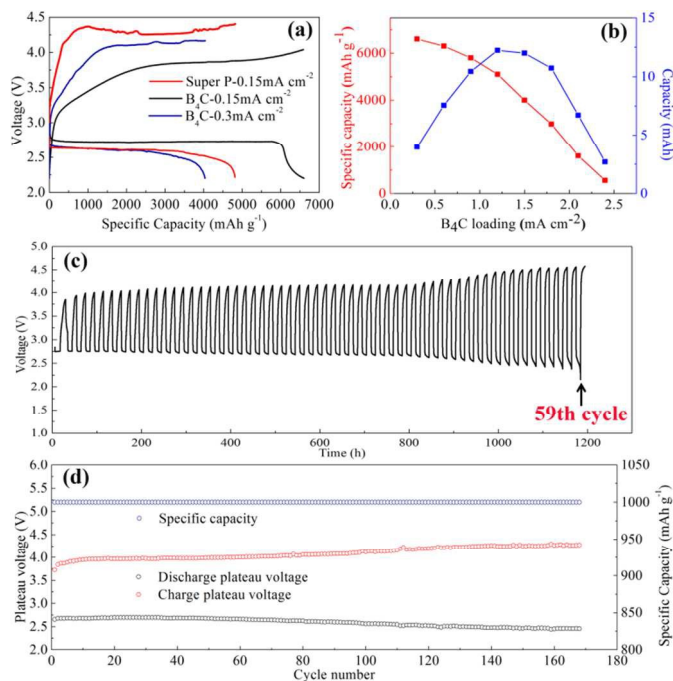


Fig. 2 (a) The initial discharge/charge curves of B₄C and Super P based Li-O₂ batteries; (b) capacity and specific capacity versus B₄C loading for the Li-O₂ batteries at 500 mA g⁻¹ (0.15 mA cm⁻²); (c) discharge/charge curves of the B₄C based Li-O₂ batteries with a cut-off capacity of 5000 mAh g⁻¹ (1.5 mAh cm⁻²), at a current density of 500 mA g⁻¹ (0.15 mA cm⁻²); (d) Plateau voltage vs. cycle number for B₄C based Li-O₂ batteries with a cut-off capacity of 1000 mAh g⁻¹ (0.3 mAh cm⁻²), at a current density of 1000 mA g⁻¹ (0.3 mA cm⁻²).

that on Super P cathode, and there is no obvious increase in R_{ct} for the B₄C based Li-O₂ battery after charging as compared with that of the pristine batteries (~420 ohm), indicating fast kinetics and reversible reaction for formation and decomposition of Li₂O₂ during the discharge/charge process on the B₄C cathode. Fig. 2b shows the plots of capacity and specific capacity versus B₄C loading for the Li-O₂ battery at a constant current density of 0.15 mA cm⁻². Although the specific capacity of the cells decreases monotonically with increasing B₄C loading, a maximum capacity of 12.24 mAh is found at B₄C loading of 1.2 mg cm⁻². When the B₄C loading is less than 1.2 mg cm⁻², the capacity decreases because less active material per unit area is used. When the B₄C loading exceeds 1.2 mg cm⁻², the capacity also decreases because a thicker electrode leads to a lower utilization of the B₄C.^[16,17]

As a consequence of reduction of the overpotential by using B₄C based electrode, the cell cyclability was maintained up to 59 cycles at current density of 500 mA g⁻¹ (0.15 mA cm⁻²) with cut-off capacity of 5000 mAh g⁻¹ (1.5 mAh cm⁻²) (Fig. 2c). It means that the overall cycling time of B₄C based Li-O₂ battery was nearly 1200 h (50 days). The cyclability of Li-O₂ battery with B₄C based electrode has been further investigated by controlling cut-off capacity of 1000 mAh g⁻¹ (0.3 mAh cm⁻²) at current density of 1000 mA g⁻¹ (0.3 mA cm⁻²) (Fig. 2d). It can be seen that Li-O₂ battery with B₄C based cathode can be discharged and charged for over 160 cycles with stable discharge and charge platform. The initial discharge and charge plateau voltage are 2.7 V and 3.7 V, respectively, after 168 cycles the plateau voltage for discharge and charge process are 2.5 V and 4.3 V, respectively. The discharge-charge voltage gap only increases 0.8 V after 168 cycles discharge and charge process, which

is far superior to the Super P electrode (Fig. S8). This is closely related to the excellent activity and stability of B₄C as a catalyst towards the reversible formation/decomposition of Li₂O₂. Such performance makes B₄C as one of the best carbon-free catalysts in Li-O₂ batteries (Tab. S3).

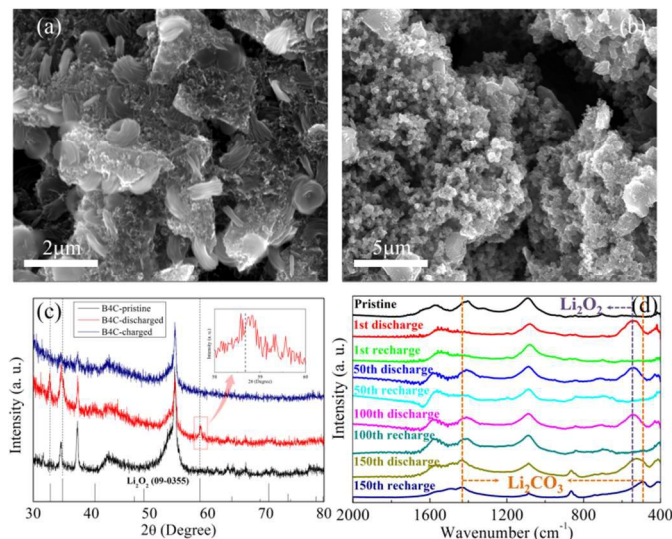
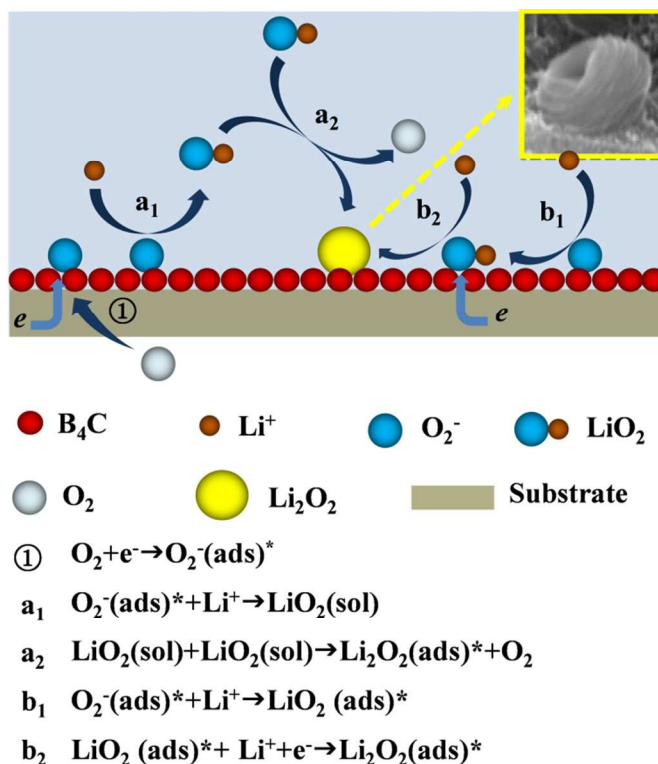


Fig. 3 SEM images of (a) discharged and (b) recharged B₄C cathode; (c) XRD patterns and (d) FTIR spectra of B₄C based electrode after different discharge/charge cycles.

To further understand the reaction mechanism over the whole process, structure and morphology evolution of the oxygen electrode at different discharge/charge stages are studied. According to the SEM images (Fig. 3 a and b), the toroidal-like Li₂O₂ (~800 nm in diameter), which consists of stacked platelets precipitates on the surface of the cathode after discharge. The rough surface of toroidal particles implied a low crystallinity of Li₂O₂ formed on B₄C cathodes. Previous results shown that this stacked toroid-like anisotropic crystallites is formed by plate-by-plate growth along the [001] direction.^[18-20] After full charging, the porous structure is regained for the B₄C electrode, indicating the superior activity of B₄C in catalysing the oxidation of Li₂O₂ during the recharging process. XRD patterns in Fig. 3c also confirm the reversible formation and decomposition of Li₂O₂ during the discharge and charge processes. The deviation of the peak corresponding to the (110) reflection is due to the different c-lattice parameter between Li₂O₂ formed under electrochemical conditions and bulk Li₂O₂ (standard reference pattern).^[18] FTIR spectrum on B₄C electrode (Fig. 3d) also demonstrates the formation and decomposition of Li₂O₂ even after 100 cycles, indicating the perfect reversibility of Li-O₂ battery using B₄C electrode. However, there are some byproducts (carbonate salt) accumulated on electrode after 150 cycles. The carbonate formed on the B₄C electrode and cannot be decomposed during the next charging step. The accumulation of byproduct during cycling has negative impact on the stability and cyclability of Li-O₂ battery.

Scheme 1 outlines the discharge mechanism based on our experimental results. O₂ is firstly reduced to O₂⁻ via one electron reduction on the B₄C (①). Based on Pearson's hard soft acid base (HSAB) theory proposed by Abraham in Li-O₂ battery,^[21,22] the basic O₂⁻ can be stabilized by the acidic surface of B₄C (the acidity of B₄C was confirmed and characterized by temperature-programmed desorption (TPD) of ammonia, Figure S9). Then O₂⁻ reacts with Li⁺ forming both solvated LiO₂ (a1) and absorbed LiO₂



Scheme 1 Illustration of discharge mechanism at B₄C surface.

(b1). Nazar believes that the nature of LiO₂ (solvated or absorbed) depends on discharge condition (e. g., current density).^[23] The succeeding predominant pathways are derived from either the further electrochemical reduction of LiO₂ or their chemical disproportionation via a solution phase or a surface migration route that forms Li₂O₂, depending on the nature of LiO₂ (solvated or absorbed). The solvated LiO₂ chemically disproportionates to Li₂O₂ (a2). When the concentration of Li₂O₂ exceeds the solubility limit, crystallites precipitate on the surface of the electrode at nucleation sites. The absorbed LiO₂ on the surface can undergo further electrochemical reduction (b2) to form Li₂O₂ via surface migration route. The acidity of B₄C is critical to stabilize aggressive O₂⁻ and inhibit side reaction originated by O₂⁻ and finally improve the stability of Li-O₂ battery.

Conclusions

In summary, compared with the traditional Super P carbon electrode, the Li-O₂ batteries using B₄C based electrode exhibit a much lower polarization, larger capacity and longer cycling life. Under oxygen condition, the cell delivers a high capacity of 6621 mAh g⁻¹ (~2 mAh cm⁻²) with narrow discharge/charge voltage gap of 1.11V at 500mA g⁻¹ (0.15 mA cm⁻²). With a cut-off capacity of 1000 mAh g⁻¹ (0.3 mAh cm⁻²) at 1000 mA g⁻¹ (0.3 mA cm⁻²), the Li-O₂ battery with B₄C electrode can be discharged and discharged over 160 cycles. The possible oxygen electrode reaction mechanism is proposed on the basis of morphology and structure investigation. Further understanding of the relationship between morphology and component of products and surface chemistry of cathode is undertaking in our lab.

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